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Interaction of feeding frequency and feeding rate on growth, nutrient utilization, and plasma metabolites of juvenile genetically improved farmed Nile tilapia, *Oreochromis niloticus*

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Abstract

We evaluated the possible interaction between feeding frequency (FF) and feeding rate (FR) on growth, nutrient utilization, body composition, and plasma metabolites of juvenile Nile tilapia, *Oreochromis niloticus* (6.57 g). We tested three FFs: two, four, and six times a day, and two FRs: apparent satiation (AS) or restricted regime (RR), following a completely randomized design in a factorial arrangement, in quadruplicate. The daily weight gain and specific growth rate were 56 and 19% higher in fish fed to AS than in fish fed under the RR, respectively. However, feed efficiency was 14% higher in fish fed in the RR. Fish fed to AS presented the highest body lipid (8.92%), whereas the fish fed in the RR showed the lowest (6.64%). Glucose and triglycerides were also higher in fish fed to AS. The protein retention ratio was highest in the RR (49.17% vs. 42.51%, respectively). Fish fed to AS showed 34% more body protein but 62% more body fat than the fish fed in the RR. There was no difference in feed efficiency or weight gain regarding FF, regardless of the

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FR. Therefore, feeding Nile tilapia juveniles twice a day is enough to achieve optimum growth.

KEYWORDS
feeding management, feeding rate, fish nutrition, *Oreochromis niloticus*

1 | INTRODUCTION

Nile tilapia (*Oreochromis niloticus*) is the third most farmed fish species worldwide, after the cyprinids (FAO, 2021). Brazil is the fourth highest Nile tilapia producer worldwide, and its production represents more than 60% of the total fish produced in Brazil (PEIXE-BR, 2021). The success of tilapia production in Brazil and other countries is largely due to advancements in genetics, nutrition, feed production, and farming systems (Michelato et al., 2016). In addition, tilapia can consume artificial feed from the first feeding stage (Boonanuntanasarn et al., 2018), promoting easy commercialization of this species. However, increasing feed and production costs are a big challenge that affects the production economics of tilapia and other farmed species. Optimizing feed and feeding management is crucial for enhancing nutrient and feed utilization, reducing production costs, and improving the profitability of tilapia production.

While feeding increases the growth and production of fish, excess feeding can reduce water quality and nutrient utilization by fish (Xu et al., 2017). Therefore, once the diets are optimized according to the nutritional requirements of the fish, feeding strategy plays a major role in improving feed efficiency and fish growth (Biswas, Thirunavukkarasu, Sundaray, & Kailasam, 2010).

Fish growth and feed utilization are generally improved when fish are fed multiple times a day with a higher feeding frequency (FF) (Dwyer, Brown, Parrish, & Lall, 2002; Tsevis, Klaoudatos, & Conides, 1992; Xie, Qinghui, Mai, Xu, & Ma, 2011). However, if the interval between meals is short, the feed can also pass through the gut quickly, resulting in less efficient nutrient absorption and poor feed utilization (Liu & Liao, 1999). A high FF also leads to a high labor cost (Wang, Kong, Li, & Bureau, 2007). On the other hand, feeding fish less frequently can also lead to reduced fish growth (Tian et al., 2015). FF is correlated with gastric evacuation time (Lee, Hwang, & Cho, 2000), and the return of appetite depends on emptying from the digestive tract (Huebner & Langton, 1982; Lee et al., 2000). The feed intake depends on the quantity of feed that remains in the stomach from the last meal. Feeding fish at intervals less than the return of the appetite can lead to an intestinal overload, resulting in less efficient nutrient absorption (Riche, Haley, Oetker, Garbrecht, & Garling, 2004).

There are evidences that under restricted feeding rate (FR), increased FF resulted in improved growth performance and feed or protein utilization in Nile tilapia (Daudpota et al., 2016; Ferdous, Nahar, Hossen, Sumi, & Ali, 2014; Garcia & Villarroel, 2009; Huang et al., 2015; Pouomogne & Ombredane, 2001; Santos, Calumby, Coelho Filho, Soares, & Gentelini, 2015; Sousa et al., 2012; Villarroel, Alavrino, & Lopez-Luna, 2011). However, under satiation feeding, increased FF resulted in poor feed utilization because of gastric overload (Riche et al., 2004). Such previous studies evaluated FF largely with restricted feeding, and no study was carried out to compare and optimize FF when different FRs are adopted. Therefore, our study was designed to test simultaneously three FFs (2, 4, and 6) when using two FRs (satiation versus restricted). We report feed intake and utilization as well as body weight for every 10–15 days over the study period, which gives close monitoring and better understanding of changes over time—a step needed to implement precision feeding. In addition, we measured serum biochemical indexes reflecting changes in lipid, protein, and carbohydrate metabolism (Yuan et al., 2019), which may help evaluate and define the ideal feed management.

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2 | MATERIALS AND METHODS

2.1 | Experimental diet

An experimental diet was formulated to meet the nutritional requirements of Nile tilapia *Oreochromis niloticus* juveniles (Furuya, 2010; NRC, 2011; Evonik's AMINOTilapia®) (Table 1).

Before diet preparation, soybean meal, corn, poultry by-product meal, and meat meal were analyzed to determine their proximate composition. The dry ingredients were ground (1-mm screen mesh, hammer mill), sieved (0.6 mm, hand sieve), weighed, and homogenized in a horizontal mixer (Inbramaq, Riberão Preto, Brazil). The oil was then incorporated and homogenized in a mixture. Moisture was adjusted to 21% with distilled water for extrusion in a single-screw extruder Model MX40 (Inbramaq, Ribeirão Preto, Brazil). Extrusion parameters were as follows: temperature 100°C, flow rate at 20% of nominal capacity, width/diameter ratio of 2.3/1, screw diameter 92.5 mm, and cylinder length 210 mm. After extrusion, pellets (1.5 mm to 2 mm) were dried in an air-circulation oven at 50°C for 4 hr. After being oven-dried, the experimental diet was packaged and stored in hermetically sealed containers, protected from light, at 4°C.

TABLE 1 Formulation and composition of the experimental diet

Ingredients	%
Soybean meal[a]	50.40
Corn[a]	33.30
Poultry by-product meal[b]	7.50
Meat meal[a]	6.40
Mineral and vitamin premix[c]	1.00
Soybean oil[d]	0.50
DL-Methionine[e]	0.35
Choline[c]	0.10
BHT[c]	0.10
Analyzed composition, % dry matter	
Crude energy (kcal/kg)	4,450
Crude protein	35.14
Dry matter	10.78
Ether extract	6.48
Ash	6.14

[a]Corn (8% crude protein), soybean meal (46% crude protein), and meat meal (49% crude protein) were provided by Nicoluzzi Rações Ltda (Penha, Brazil).
[b]Poultry by-product meal (65% crude protein), produced by Kabsa S.A. (Porto Alegre, Brazil).
[c]Choline bitartrate (1.0 g/kg), butylated hydroxytoluene (BHT) (1.0 g/kg) and vitamin-micromineral premix (produced by Cargill, São Paulo, Brazil), composition per kg: folic acid 420 mg, pantothenic acid 8,333 mg, BHT 25,000 mg, biotin 134 mg, cobalt sulfate 27 mg, copper sulfate 1833 mg, iron sulfate 8,000 mg, calcium iodate 92 mg, manganese sulfate 3,500 mg, niacin 8.333 mg, selenite 100 mg, vitamin (vit.) A 1,666,670 UI, vit. B_1 2083 mg, vit. B_{12} 5,000 μg, vit. B_2 4,166 mg vit. $B_6$3166 mg, ascorbic acid equivalent 66,670 mg, vit. D_3 666,670 UI, vit. E 16,666 UI, vit. K_3 833 mg, zinc sulfate 23,330 mg, inositol 50,000 mg, and calcium propionate 250,000 mg.
[d]Soybean oil (produced by Bunge Alimentos S.A., Gaspar, Brazil).
[e]DL-Methionine produced by Evonik Industries AG (Essen, Germany).

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2.2 | Feeding protocol

The feeding trial was carried out at 27.54°C, following a completely randomized design, in a factorial arrangement. Two factors were tested: FF (two, four, or six times a day) and FR (restricted or apparent satiation (AS)), totaling six dietary treatments. Each treatment was tested in quadruplicate groups of tilapia at 26 fish per tank (Table 2). The FR for the restricted regime (RR) was set as 85% of the mean feed intake (% body weight) of fish fed to apparent satiation (twice daily). This FR was adjusted every 12 days after biometrics. In case of mortality at any given time, the feed offered to the tank was adjusted to the number of live fish. Fish were hand fed to prevent any feed wastage which could hinder feed efficiency calculation mainly in the satiation feeding regime. Feeding to AS was accomplished by carefully monitoring feeding activity and ceasing to offer feed when fish stop ingesting the pellets.

2.3 | Experimental procedures

The feeding trial was run at the Laboratório de Biologia e Cultivo de Peixes de Água Doce (LAPAD-UFSC), Brazil. Juvenile Nile tilapia of the genetically improved farmed tilapia (GIFT) strain were obtained from a local fish farm (Piscicultura Pomerode, Pomerode, SC, Brazil) after masculinization with 17-alpha-methyltestosterone (>99%). Fish handling followed the protocol number 9377080618, approved by the Ethics Committee on Animal Use of the Universidade Federal de Santa Catarina (CEUA, UFSC).

Fish were acclimated to the experimental conditions for 14 days. For the first 7 days, fish were stocked in three 1,000-L tanks, and, for the remaining 7 days, they were transferred into 24 tanks (six treatments × four replicates). Each 100-L tank (experimental unit) was connected to the same recirculating aquaculture system, with a controlled water temperature of 27.54°C as well as mechanical and biological filtration. Groups of 26 juvenile Nile tilapia (6.57 ± 0.10 g, mean body weight ± SD) were randomly distributed into each tank. During the acclimation period, fish were fed the experimental diet daily to AS at 10:00 and 17:30 hr. The photoperiod was adjusted to 12 h.

2.4 | Fish biometrics and sample collection

At the beginning and end of the feeding trial, which lasted 51 days, fish were individually measured for total length and body weight. Fish weight was also recorded every 10 or 14 days to monitor growth. There was a total of four feeding periods: first period (Day 1 to Day 10), second period (Day 11 to Day 25), third period (Day 26 to Day 36), and fourth period (Day 37 to Day 51). Before biometrics, all fish were fasted for 24 hr and anesthetized with

TABLE 2 Fish feeding protocol

Feeding frequency	Ration level	Feeding time (hr)
2	Apparent satiation	10:00, 17:30
2	Restricted	10:00, 17:30
4	Apparent satiation	10:00, 12:30, 15:00, 17:30
4	Restricted	10:00, 12:30, 15:00, 17:30
6	Apparent satiation	10:00, 11:30, 13:00, 14:30,16:00, 17:30
6	Restricted	10:00, 11:30, 13:00, 14:30, 16:00, 17:30

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100 mg/L Eugenol® (Biodinâmica Química and Farmacêutica Ltd., Ibiporã, Brazil). The following variables were calculated using the fish biometrics and feed consumption data:

$$\text{Weight gain WG, g} = (\text{Final weight} - \text{initial weight})$$

$$\text{Daily weight gain DWG, g/day} = \left(\text{Weight gain} \times \text{day}^{-1}\right)$$

$$\text{Specific growth rate SGR, \%/day} = \left[\{\ln \text{final weight} - \ln \text{initial weight}\} \times \text{day}^{-1}\right] \times 100$$

$$\text{Feed efficiency (FE)} = \left[(\text{Weight gain}) \times (\text{total feed intake})^{-1}\right] \text{x}100$$

$$\text{Protein retention rate (PRR), \%} = [\text{Final weight} \times \text{final body protein}] - [\text{initial body weight} \times \text{initial body protein}] \times [\text{protein intake}]^{-1}$$

$$\text{Lipid retention rate (LRR), \%} = [\text{Final weight} \times \text{final body lipid}] - [\text{initial body weight} \times \text{initial body lipid}] \times [\text{lipid intake}]^{-1}$$

Survival was monitored daily in each tank. In each experimental unit, feed intake was carefully measured by weighing each feed-storage container at the end of each day.

For the whole-body proximate composition analysis, three groups of 20 fish each were sampled at the beginning of the experiment, and three fish per experimental unit were sampled at the end of the experiment. Prior to sampling, fish were fasted for 24 hr and euthanized via overdose (200 mg/L) of the anesthetic Eugenol®, followed by sectioning of the spine. The collected samples were dried, homogenized, and stored at −20°C until analysis.

Blood samples were collected from the caudal vein of three anesthetized fish from each tank for plasma-metabolite determination.

2.5 | Proximate analysis

The proximate analysis was conducted in the Fish Nutrition Lab (LABNUTRI, UFSC), and the following procedures were standardized by the Association of Official Analytical Chemists (AOAC, 1999): moisture (drying at 105°C to constant weight, method 950.01), crude protein (Kjeldahl, method 945.01), total lipid (Soxhlet, method 920.39C), and ash (incineration at 550°C, method 942.05). Crude energy was determined in a calorimeter bomb (PARR, model ASSY 6200, Moline, IL) according to the manufacturer's instructions. Three samples (100 g) from the experimental diet were also analyzed following these procedures.

2.6 | Plasma metabolites

After the feeding trial, immediately after blood sampling, an aliquot of blood was placed on a reagent strip to determine glucose (Glu), using a digital glucosimeter One Touch® Ultra® (Johnson & Johnson Medical, São Paulo, Brazil). Plasma was obtained after blood centrifugation (6,000 rpm for 15 min) and stored at − 20°C until analysis. Quantitative determination of plasma metabolites, total proteins (TP), triglycerides (TG), and cholesterol (CHL) was carried out by enzymatic colorimetric analysis with the use of commercial kits (Labtest Diagnóstica SA, Lagoa Santa, Brazil: Cholesterol Ref. 76; Triglycerides Ref. 87; Total protein Ref. 99).

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2.7 | Water quality parameters

Water quality parameters were measured weekly, except temperature, dissolved oxygen, and pH, which were monitored daily. The average values (± standard deviation) were as follows: temperature, 27.54 ± 0.86°C; dissolved oxygen, 5.05 ± 0.77 mg/L; pH 6.98 ± 0.36; salinity 1.48 ± 0.22 g/L; alkalinity 58.92 ± 8.03 $CaCO_3$ mg/L; total ammonia 0.55 ± 0.10 mg/L, and 0.01 mg/Lnitrite. All measured variables remained within the comfort range for Nile tilapia (Popma & Lovshin, 1995). The water inflow rate in each experimental unit was 25 mL/s.

2.8 | Statistical analyses

All data were first analyzed for normality (Shapiro–Wilk) and homoscedasticity (Levene). Data were analyzed in SAS using the Proc mixed model (two-way ANOVA), which included FF, ration level, and their interaction as the fixed factors and tank as the random factor. The mean test (Tukey's test) was performed only when significant differences were detected. The level of significance adopted in all tests was 5%.

Data on growth performance, feed intake, and feed efficiency were analyzed considering each of the four feeding periods of 12 days between fish biometrics. In addition, data on growth performance, feed intake, feed efficiency, and protein and lipid retention were also analyzed considering the entire feeding period of 51 days.

During the first week of the feeding trial, some fish deaths occurred, especially in one tank (experimental unit) due to fish agonistic behavior. Data outliers were removed when analyzing weight gain and feed intake from the tank where fish were fed to AS four times a day.

3 | RESULTS

3.1 | Growth, feed intake, and feed efficiency over the study period

There was no significant interaction between FF and FR with respect to weight gain in all periods (Tables 3 and 4). Weight gain was highly affected by the feeding rate in all periods but was not affected by FF. Weight gain of fish fed to AS was significantly higher than that of fish fed in the RR, regardless of the analyzed feeding period (Figure 1).

The specific growth rate was highly affected by the FR in all periods ($p < .05$; Tables 3 and 4), but it was not affected by FF ($p > .05$). There was a marginal interaction effect ($p = .08$) between FF and FR on the specific growth rate in the first (Days 1–10) and third (Days 26–36) feeding periods. In the first period, the fish groups fed to AS four and six times a day had a marginal increase in specific growth rate compared to fish fed twice a day (Table 3). However, a marginal decrease in specific growth rate was evident in fish groups fed to AS four and six times a day when compared to fish fed twice a day during the third period (Days 26–36) (Table 4).

There was a significant interaction between FF and FR on feed intake in the first (Days 1–10) and second (Days 11–25) periods (Table 3). The feed intake increased as FF increased from two to four or six times a day, but only in fish fed to AS. Feed intake was greatly affected by FR in all periods and was highest in fish groups fed to AS compared to fish groups fed in the RR (Tables 3 and 4). Feed intake was not affected by FF in the third (Days 26–36) and fourth (Days 37–51) periods. In addition, there was no interaction between FF and FR on feed intake during the third and fourth periods (Table 4).

Feed efficiency was significantly affected by FR and frequency in fish fed to AS. During the first period (Days 1–10), fish fed twice a day showed a higher feed efficiency than fish fed four or six times a day. However, there was no interaction between FR and frequency during the first period (Table 3). During the second, third, and fourth periods, fish fed in the RR presented a higher feed efficiency than fish fed to AS (Tables 3 and 4).

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TABLE 3 Average weight gain, specific growth rate, feed intake, and feed efficiency of Nile tilapia juveniles fed at different feeding rates and feeding frequencies in the first and second feeding periods

Feeding rate	Feeding frequency	Weight gain, g	SGR, %	Feed intake, g/fish	Feed efficiency, %	Feeding rate	Feeding frequency	Weight gain, g	SGR, %	Feed intake, g/fish	Feed efficiency, %
First period (Days 1 to 10)						**Second period (Days 11 to 25)**					
Restricted	**2**	4.51	4.78	3.83^{x}	113.00^{A}	**Restricted**	**2**	9.97	4.58	9.36^{x}	105.25
	4	4.49	4.76	3.83^{x}	113.00^{B}		**4**	10.07	4.62	9.40^{x}	106.00
	6	4.45	4.72	3.83^{x}	113.00^{B}		**6**	10.07	4.62	9.39^{x}	106.00
	Mean	*4.48*^{b}	*4.75*^{b}	*3.83*	*113.00*^{a}		**Mean**	*10.04*^{b}	*4.60*^{b}	*9.38*	*105.75*^{a}
Apparent satiation	**2**	6.81	6.50	6.52^{y}	105.00^{A}	**Apparent satiation**	**2**	17.35	5.95	18.14^{y}	95.00
	4	6.30	6.89	7.50^{z}	98.00^{B}		**4**	18.04	5.98	19.62^{z}	90.00
	6	7.31	6.92	7.56^{z}	98.00^{B}		**6**	18.08	6.00	19.25^{z}	93.00
	Mean	*7.14*^{a}	*6.77*^{a}	7.19	*100.33*^{b}		**Mean**	*17.82*^{a}	*5.98*^{a}	*19.00*	*92.67*^{b}
	Pooled SEM	6.80	7.22	7.07	12.92		**Pooled SEM**	8.72	7.42	9.82	11.92
Probability value—two-way ANOVA						**Probability value—two-way ANOVA**					
Feeding rate (FR)		<0.01	<0.01	<0.01	<0.01	Feeding rate (FR)		<0.01	<0.01	<0.01	<0.01
Feeding frequency (FF)		0.27	0.19	<0.01	0.05	Feeding frequency (FF)		0.47	0.83	<0.01	0.58
FR x FF		0.19	0.08	<0.01	0.13	FR x FF		0.64	0.99	<0.01	0.36

Note: Initial weight: 6.57 ± 0.1 g (mean ± standard deviation). Superscript lowercase letters (a,b,c) were given to separate means when there was a significant difference between feeding rates (restricted vs. satiation). Superscript uppercase letters (A,B,C) were given to separate means when there was a significant difference among feeding frequencies (two, four, or six times a day). Superscript lowercase letters (x,y,z) were given to separate means when there was a significant interaction between feeding rate and feeding frequency.
Abbreviations: ANOVA, analysis of variance; SEM, Standard error of means; SGR, specific growth rate.

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TABLE 4 Average weight gain, specific growth rate, feed intake, and feed efficiency of Nile tilapia juveniles fed at different feeding rates and feeding frequencies in the third and fourth feeding periods

Feeding rate	Feeding frequency	Weight gain, g	SGR, %	Feed intake, g/fish	Feed efficiency, %	Feeding rate	Feeding frequency	Weight gain, g	SGR, %	Feed intake, g/fish	Feed efficiency, %
Third period (Days 26 to 36)						**Fourth period (Days 37 to 51)**					
Restricted	**2**	11.99	4.54	11.77	103.00	**Restricted**	**2**	18.62	3.00	20.67	89.00
	4	12.84	4.79	11.81	109.00		**4**	18.73	3.10	20.66	90.00
	6	12.21	4.60	11.82	105.00		**6**	19.80	2.95	20.67	96.00
	Mean	12.35[b]	4.64[b]	11.80[b]	105.67[a]		**Mean**	19.05[b]	3.02[b]	20.67[b]	91.67[a]
Apparent satiation	**2**	19.81	5.00	21.30	93.00	**Apparent satiation**	**2**	25.76	3.29	30.98	80.00
	4	19.66	4.83	21.50	91.00		**4**	27.17	3.23	31.93	83.00
	6	19.92	4.88	21.45	93.00		**6**	25.88	3.43	31.08	80.00
	Mean	19.79[a]	4.90[a]	21.42[a]	92.33[b]		**Mean**	26.27[a]	3.31[a]	31.33[a]	81.00[b]
	Pooled SEM	8.54	3.18	11.25	9.76		**Pooled SEM**	7.13	2.34	9.77	13.17
Probability value—two-way ANOVA						**Probability value—two-way ANOVA**					
Feeding rate (FR)		<0.01	<0.01	<0.01	<0.01	Feeding rate (FR)		<0.01	<0.01	<0.01	<0.01
Feeding frequency (FF)		0.79	0.96	0.71	0.47	Feeding frequency (FF)		0.53	0.86	0.45	0.59
FR x FF		0.56	0.98	0.08	0.08	FR x FF		0.28	0.15	0.44	0.33

Note: Superscript lowercase letters (a,b,c) were given to separate means when there was a significant difference between feeding rates (restricted vs. satiation).
Abbreviations: ANOVA, analysis of variance; SEM, Standard error of means; SGR, specific growth rate.

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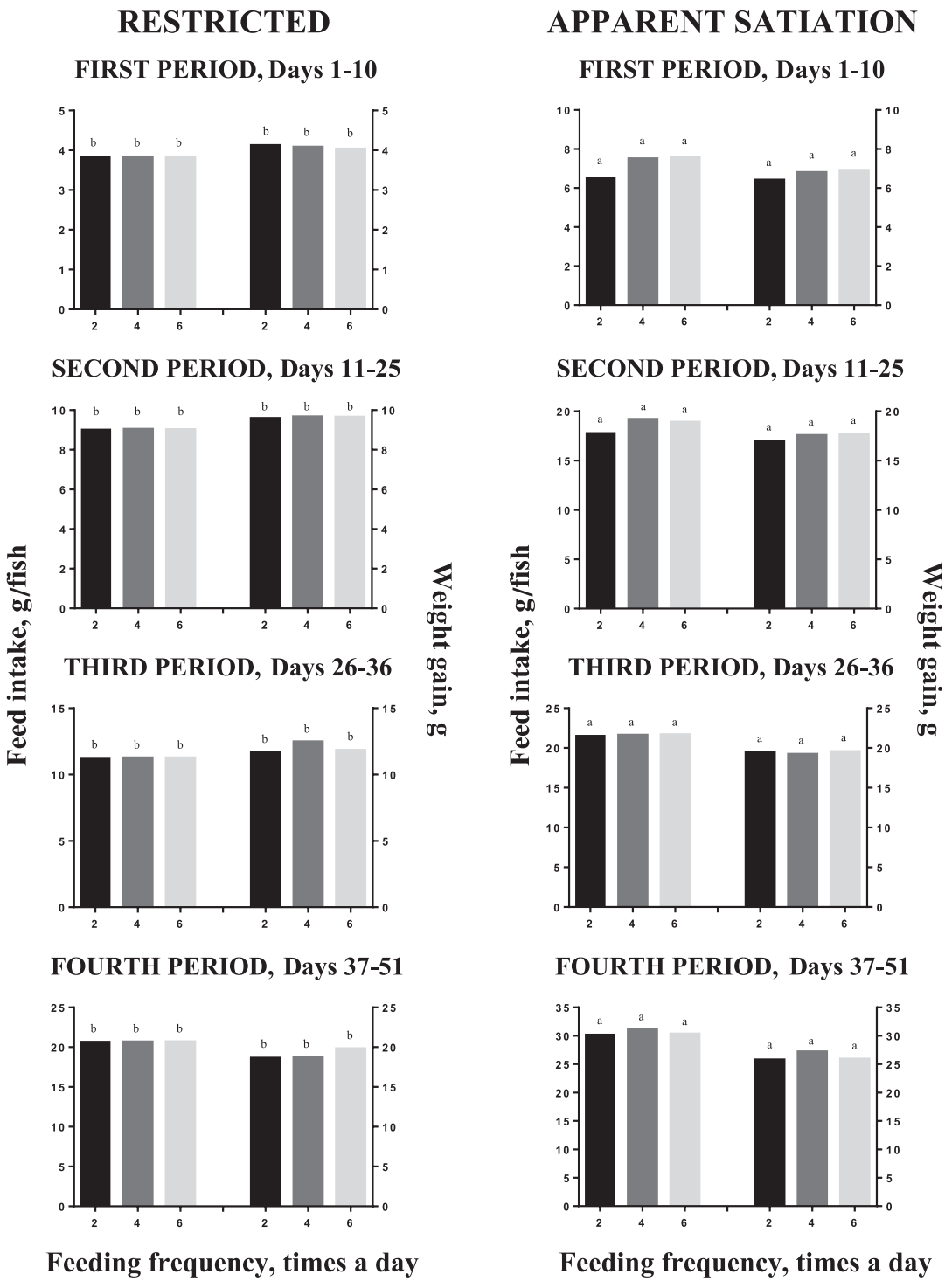


FIGURE 1 Feed intake and weight gain of Nile tilapia juveniles within each feeding rate (restricted versus apparent satiation) and feeding frequency (two, four, or six times a day) at different time spans along the experimental period. [a,b] To separate means within each feeding rate or feeding frequency

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TABLE 5 Average growth, feed intake and feed efficiency, protein and lipid retention rate of Nile tilapia juveniles fed at different feeding strategies for 51 days

Feeding rate	Feeding frequency	Final weight, g	Daily weight gain, g/day	Specific growth rate, %	Feed intake, g/fish	Feed efficiency	Protein retention rate, %	Lipid retention rate, %
Restricted	**2**	50.22	0.86	4.07	45.43^{x}	0.97	48.87	103.26
	4	51.50	0.88	4.11	45.50^{x}	1.00	49.70	100.36
	6	51.90	0.89	4.12	45.50^{x}	1.00	48.93	103.58
	Mean	*51.21*^{b}	*0.88*^{b}	*4.10*^{b}	*45.48*	*0.99*^{a}	*49.17*^{a}	*102.40*^{b}
Apparent satiation	**2**	75.19	1.34	4.83	76.74^{y}	0.89	42.90	122.62
	4	77.30	1.39	4.92	80.31^{z}	0.85	42.26	123.48
	6	76.52	1.37	4.90	79.14^{zy}	0.88	42.38	131.34
	Mean	*76.34*^{a}	*1.37*^{a}	*4.88*^{a}	*78.72*	*0.87*^{b}	*42.51*^{b}	*125.81*^{a}
Pooled SEM		*3.16*	*0.06*	*0.09*	*0.71*	*0.04*	*1.89*	*8.42*
Probability value—two-way ANOVA								
Feeding rate		<0.01	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
Feeding frequency		0.44	0.43	0.06	0.04	0.71	0.91	0.26
Feeding rate x feeding frequency		0.88	0.88	0.88	0.05	0.13	0.64	0.49

Note: Initial weight: 6.57 ± 0.1 g (mean ± standard deviation). Superscript lowercase letters (a,b,c) were given to separate means when there was a significant difference between feeding rates (restricted vs. satiation). Superscript lowercase letters (x,y,z) were given to separate means when there was a significant interaction between feeding rate and feeding frequency. Abbreviation: ANOVA, analysis of variance; SEM, standard error of means.

3.2 | Growth variables, survival, feed intake, feed efficiency, and nutrient retention considering the whole feeding trial

The average fish survival rate was 95.83% and was not affected by FF or FR. The average survival for each dietary treatment was as follows: restricted-twice, 94.23%; restricted-four, 97.12%; restricted-six, 94.23%; satiation-twice, 99.04%; satiation-four, 90.38%; and satiation-six, 100%. The few registered mortalities occurred during the first week of the trial and were due to the agonistic behavior shown by the GIFT lineage. One tank (experimental unit) where fish were fed four times a day to AS was especially affected (73% fish survival rate).

The growth variables, feed intake, feed efficiency, and nutrient retention considering the whole period of the feeding trial are presented in Table 5. There was no interaction between FF and FR regarding final weight, daily weight gain, specific growth rate, or feed efficiency. FR affected all these variables: fish fed to AS always presented the highest values, regardless of FF. FF marginally affected ($p = .06$) fish-specific growth rate: fish fed twice a day had the lowest specific growth rate, regardless of the FR. There was a significant interaction between FF and FR regarding feed intake: fish fed to AS, four times a day, presented the highest feed intake, whereas fish fed in the RR had the lowest feed intake, regardless of the FF. The nutrient retention rate was only affected by FR, but protein retention reached the highest value in fish fed in the RR, whereas lipid retention was the highest in fish fed to AS. There was no interaction between FF and FR with respect to the nutrient retention rate.

3.3 | Body composition

An interaction was evident between FF and FR regarding body moisture and body protein composition (Table 6). Body protein was highest in fish fed two or four times a day in the restricted feed regime compared to fish fed twice a day to AS. Fish fed to AS accumulated more body fat but less ash content than fish fed in the RR, regardless of FF.

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TABLE 6 Average body proximate composition (g/kg wet weight) of Nile tilapia juveniles fed at different feeding strategies for 51 days

Feeding rate	Feeding frequency	Moisture %	Crude protein	Ether extract	Ash
Restricted	**2**	72.77^{xy}	17.13^{x}	6.74	3.70
	4	72.82^{xy}	17.13^{x}	6.52	3.72
	6	73.25^{x}	16.81^{xy}	6.67	3.55
	Mean	*72.94*	*17.02*	*6.64*^{a}	*3.66*^{b}
Apparent satiation	**2**	72.21^{zy}	16.48^{y}	8.58	3.20
	4	71.31^{z}	16.80^{xy}	8.86	3.39
	6	71.23^{z}	16.89^{xy}	9.31	3.37
	Mean	*71.58*	*16.72*	*8.92*^{b}	*3.32*^{a}
Pooled SEM		*0.55*	*0.31*	*0.51*	*0.23*
Probability value—two-way ANOVA					
Feeding rate		<0.01	<0.01	<0.01	<0.01
Feeding frequency		0.19	0.43	0.30	0.47
Feeding rate x feeding frequency		0.01	0.04	0.24	0.25

Note: Superscript lowercase letters (a,b,c) were given to separate means when there was a significant difference between feeding rates (restricted vs. satiation). Superscript lowercase letters (x,y,z) were given to separate means when there was a significant interaction between feeding rate and feeding frequency.
Abbreviation: SEM, standard error of means.

TABLE 7 Average plasma metabolites of Nile tilapia juveniles fed at different feeding frequencies and feeding rates for 51 days

Feeding rate	Feeding frequency	Total protein g/dL	Glucose mg/dL	Cholesterol mg/dL	Triglycerides mg/dL
Restricted	**2**	3.03	54.83	106.86	90.95
	4	3.46	54.42	110.46	71.55
	6	3.34	58.22	95.56	66.86
	Mean	*3.28*	*55.82*^{a}	*104.29*	*76.45*^{a}
Apparent satiation	**2**	3.38	60.56	113.94	101.32
	4	3.55	61.33	113.75	98.68
	6	3.48	62.94	114.88	111.86
	Mean	*3.47*	*61.61*^{b}	*114.19*	*103.95*^{b}
Pooled SEM		*0.47*	7.12	*25.49*	*19.47*
Probability value—two-way ANOVA					
Feeding rate		0.21	0.04	0.26	<0.01
Feeding frequency		0.30	0.64	0.79	0.44
Feeding rate x feeding frequency		0.77	0.55	0.73	0.14

Note: Superscript lowercase letters (a,b,c) were given to separate means when there was a significant difference between feeding rates (restricted vs. satiation).
Abbreviations: ANOVA, analysis of variance; SEM, standard error of means.

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3.4 | Plasma metabolites

There was no interaction between FR and FF for the plasma metabolites evaluated (Table 7). Glucose and triglycerides were affected by the FR: fish fed to AS had the highest values compared to fish fed in the RR. Total protein and cholesterol were not affected by the FR or FF.

4 | DISCUSSION

The weight gain was 50% higher in fish fed to AS than in those fed under the RR. Higher FRs promote higher nutrient intake, which tends to improve growth. These findings with the omnivore Nile tilapia (GIFT strain) corroborate what has already been reported in other fish species, with different feeding habits (Arzel, Metailler, Gall, & Guillaume, 1998; Cho, Lee, Park, & Lee, 2006; Li et al., 2018; Van Ham et al., 2003; Xu et al., 2017).

The average specific growth rate of fish fed twice a day to AS was close to that reported in Nile tilapia of similar weight, FRs and frequencies, water temperature, and diet composition (control treatment) (Fernandes, Brignol, Filler, Pettigrew, & Fracalossi, 2018). Additionally, several fish species, regardless of feeding habit, have been reported to have high specific growth rates when fed at high FRs (Eroldoğan, Kumlu, & Aktas, 2004; Ng, Lu, Hashim, & Ali, 2000; Robinson & Li, 1999; Silva, Gomes, & Brandão, 2007; Sun, Chen, & Huang, 2006). Nile tilapia specific growth rate was 20% higher in fish fed to AS than in fish fed under the RR. Although no difference was evident among FFs regarding specific growth rate, fish fed to AS four and six times a day presented a marginally ($p = .08$) higher specific growth rate compared to the fish fed twice daily, considering the first feeding period. Such marginal difference is noteworthy because small fish tend to benefit from multiple daily feedings (Dwyer et al., 2002; Tsevis et al., 1992; Xie et al., 2011). However, in the third period, this trend was inverted and fish fed to AS twice a day presented marginally ($p = .08$) higher specific growth rate than fish fed four or six times a day. When considering the whole feeding period, the specific growth rate was only affected by FR: fish fed to AS presented a higher specific growth rate than those fed under the RR. However, a marginal difference ($p = .06$) was also detected in FF: fish fed two times a day had the lowest specific growth rate. Overall, our findings suggest that the specific growth rate increases when Nile tilapia is fed to AS four or six times a day, but such a feeding regime increases feed intake. Feeding fish four times a day promoted the highest feed intake compared to feeding fish two or six times daily. However, such an increase in feed intake was not followed by an increase in weight gain.

Our findings show that feeding fish to AS impairs feed efficiency. Indeed, feed efficiency was 14% higher in Nile tilapia fed under the RR. Fish tend to optimize their digestion process when fed in a RR, extracting nutrients more efficiently from the feed (Eroldoğan et al., 2004; Van Ham et al., 2003). Feed efficiency is related to feed management since the return of the appetite is related to the rate of gastric emptying (Lee et al., 2000). Gastric emptying time is a major mechanism affecting FF in fish (Silva et al., 2007; Van Der Meer, Herwaarden, & Verdegem, 1997). It was previously reported that Nile tilapia fed to AS consumed the same amount of feed, regardless of feeding three or five times a day; however, fish fed three times a day had the highest growth and feed efficiency (Riche, 2000).

The return of appetite in adult Nile tilapia (average weight 183 g) fed a pelleted diet to AS was reported to be approximately 4 hr; feeding fish at shorter intervals may result in gastric overload, which results in reduced feeding efficiency (Riche et al., 2004). However, no difference in feed efficiency was detected in our study despite the meal intervals being 1:30, 2:30, or 7:30 hr. We could speculate that, since we used an extruded diet, Nile tilapia could absorb nutrients from the diet faster and more efficiently than the pelleted diet used in the cited study (Riche et al., 2004). In addition, a difference in the time required for the return of appetite between 70-g fish (our study) when compared to 183-g fish (Riche et al., 2004) cannot be ruled out. In addition to the fish size and diet-processing method, another possible explanation for the lack of difference in daily feed intake when fish were fed to AS, regardless of FF, could be attributed to the balance between the feed retention time in the gut and the energy expended for feeding. If the time between daily feeding is short, feed quickly passes through the digestive tract, reducing the

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absorption of nutrients, as reported for the short-gut carnivores *Morone saxatilis* × *Morone chrysops* hybrid (Liu & Liao, 1999) and Asian sea bass *Lates calcarifer* (Biswas et al., 2010). However, this was not observed in our trial, since there was no difference in feed efficiency among fish fed two, four, or six times a day. We hypothesized that the high intestinal coefficient (gut length body length-1) of Nile tilapia, averaging 6.14, as previously reported by our lab (Rodrigues, Gominho-Rosa, Cargnin-Ferreira, Francisco, & Fracalossi, 2012), could compensate for the efficient absorption of nutrients, even when fed six times a day.

Body composition is often used as an indicator of fish growth quality and is affected by several factors, including FF and FR (Ahmed, 2007; Khan, Jafri, & Chadha, 2004). However, body protein is not usually affected by FR since fish fed at maintenance ration levels can still deposit body proteins while mobilizing body lipids (NRC, 2011). Although body protein was highest in fish fed two or four times a day in the restricted feed regime compared to fish fed twice a day to AS, this difference represented only a 4% increment in feed intake. There was no interaction between FF and FR with respect to protein retention rate in Nile tilapia juveniles.

One important finding was that Nile tilapia fed to AS presented 34% more body protein but 62% more body lipid than when fed under the RR. On the other hand, fish fed to AS presented the lowest body ash. Increasing the FR usually increases whole-body lipid content and reduces moisture content (Abdelghany, Ayyat, & Ahmad, 2002; Eroldoğan et al., 2004; Mihelakakis, Tsolkas, & Yoshimatsu, 2002; Ng et al., 2000; Shimeno, Shikata, Hosokawa, Masumoto, & Kheyyali, 1997). One study also reported a reduction in body ash (Van Ham et al., 2003). The same trend was found for body lipid and ash in our study, but body moisture was higher in fish fed six times a day at the restricted FR compared to fish fed to AS, regardless of FF. The surplus of feed consumed by fish fed to AS, as detected in our study, usually results in energy reserve accumulation, which could lead to an increase in body lipids (McCue, 2010).

Nile tilapia fed under the restricted feeding regime showed the highest protein retention rate, but the lowest lipid retention rate. At high FRs, fish tend to reduce protein retention (Xu et al., 2017). In this case, the excess protein is catabolized, being: 1) diverted to energy metabolism, increasing the lipid retention rate (Li et al., 2018), which results in lipid deposition (Deng et al., 2003) or 2) excreted as ammonia to the water (Steffens, 1989; Webb & Gatlin, 2003). Similar nutrient retention rates were reported in Nile tilapia (Meyer-Burgdorff, Osman, & Gunther, 1989), channel catfish (Xu et al., 2017), olive flounder (Cho et al., 2006), and white sturgeon *Acipenser transmontanus* (Deng et al., 2003). The decrease in protein retention rate can be explained by a higher energy demand for feed assimilation at higher feed intake (Meyer-Burgdorff et al., 1989), as reported earlier in Nile tilapia fed to AS. This finding was also reported in tambaqui, where fish presented the highest nutrient utilization at low FRs (Silva et al., 2007). Corroborating these findings, Nile tilapia fed under the RR utilized the dietary protein more efficiently, as shown by the high protein retention rate. We suggest that further studies regarding the restriction level should be conducted to optimize body weight gain and feed utilization.

Blood glucose and triglycerides are important indicators of lipid metabolism in fish (Xu et al., 2017). Increasing levels of circulating triglycerides might be a result of increasing body lipid composition, which also might reflect reduced rates of lipid catabolism (McCue, 2010). Indeed, fish fed to AS showed the highest blood glucose levels, blood triglycerides, body lipid content, and lipid retention rate. Similar results were reported in common carp *Cyprinus carpio*, where lipogenesis was depressed and fatty acids in the hepatopancreas were mobilized to supply energy for the maintenance metabolism in fish fed under a RR (Shimeno et al., 1997). In addition, in channel catfish, higher blood glucose, triglycerides, and lipid retention were registered in fish fed at a high FR, where more protein could be used for energy metabolism and be converted into lipids, thus promoting lower protein retention (Xu et al., 2017). Therefore, when fed to AS, fish tend to use protein less efficiently, as shown by the low protein retention, and the excess dietary protein is probably catabolized to yield energy. These findings were confirmed by the measured plasma metabolites.

5 | CONCLUSIONS

Our findings show no great value in an increased FF at either the restricted or the satiation feeding regime. There was no difference in feed efficiency or weight gain among fish fed two, four, or six times a day when fish were fed

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to AS or at a restricted feeding regime. Fish weight gain was only affected by the FR. Fish fed to AS presented the highest weight gain but lower feed efficiency, accumulating 34% more body protein but 62% more body fat than fish fed under the RR. Therefore, considering the feeding level and the time interval between meals used in this trial, feeding Nile tilapia juveniles twice a day is enough to achieve optimum growth.

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CONFLICT OF INTEREST

One of the authors (Karthik Masagounder) is an employee of Evonik, the funders, and he was involved in a committee containing both Evonik staff and staff of UFSC, where the research project was developed and approved.

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